

# JEFFREY WANG

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## EDUCATION

### Carnegie Mellon University

**MS in Computational Biology**, 4.00/4.00

Aug 2025 - Dec 2026

Pittsburgh, PA

### Northeastern University

**BS in Computer Science, Minor in Biology**, 3.67/4.00

Sep 2021 - May 2025

Boston, MA

**Coursework:** Machine Learning, Natural Language Processing, Applied Data Science, Bayesian Statistics, Genetics, Molecular Biology, Genome Editing, Cancer Immunotherapies

## EXPERIENCE

### The Broad Institute, Massachusetts General Hospital - Vazquez-Garcia Lab

Cambridge, MA

Student Researcher

Apr 2025 - Jul 2025

- Authored HPC cluster (SLURM) documentation, improving job submission consistency for a team of 5 researchers.
- **Analyzed 392,000+ cancer cells** via a single-cell RNA-seq workflow to perform cell-type clustering, annotation, and differential gene expression. Presented key findings to PI to refine analyses and identify novel gene signatures.
- Fronted the generation of preliminary figures (UMAPs, heatmaps) that were **incorporated into 3 grant applications**.

### Blueprint Medicines

Cambridge, MA

Bioinformatics Co-op

Jul 2024 - Dec 2024

- Engineered and deployed a production-grade end-to-end proteomics analysis pipeline on **AWS** using **Nextflow** and **Docker**, automating QC, dose-response modeling, and differential abundance testing.
- Developed a GSEA reporting workflow for mass spectrometry data, that **slashed analysis and visualization turnaround time by 99%** (from 1 week to 15 minutes).
- Took the initiative to create an **RShiny** app to calculate drug synergy scores, aiding lead compound prioritization.
- Performed differential expression and pathway analysis (**DESeq2**, **GSEA**) on **RNA-seq data** from 6 cell lines to elucidate mechanisms of drug resistance, presenting findings to project stakeholders.

### Verve Therapeutics

Boston, MA

Computational Biology Co-op

Jul 2023 - Dec 2023

- Developed a validation framework in **Python** and **R** to benchmark and improve CRISPR gRNA on-target efficiency models against applied **transfer learning on published LLMs** and a prototyped two-encoder model with **PyTorch**.
- Trained and optimized a sequence-feature-based **XGBoost** model on over 40,000+ gRNA samples to **predict CRISPR on-target efficiency with an  $R^2$  of 0.91**. **Reduced the experimental search space for researchers by 30%**.
- Engineered a genomics data pipeline (**BLAST**, **Ensembl APIs**) to automate the extraction of sequence and structural features, a critical prerequisite for the efficiency model.
- Drove project from model development to final validation, presenting at a company-wide research symposium.

## PROJECTS

### Machine Learning for Chronic Kidney Disease Prediction | Artificial Intelligence Project

Jan 2025 - Apr 2025

- Devised a novel data synthesis method using **Kernel Density Estimation (KDE)** and **rejection sampling** to augment training clinical data and improve model generalization on out-of-distribution samples.
- Made **Random Forest** and **Multilayer Perceptron (MLP)** classifiers from scratch in Python; the MLP, using L2 regularization and dropout, achieved **97% accuracy** on the external validation set.
- Identified clinically relevant biomarkers through permutation feature importance and presented the complete methodology, validation strategy, and findings in a 8-page formal report and research poster.

### Full Stack Q and A Web Forum | Fundamentals of SWE Project

Jan 2025 - Apr 2025

- Collaborated in an Agile team of 4 to build a full-stack Q and A web app with two-week sprints and user stories.
- Led the backend development, building the **Node.js/Express** server and **MongoDB** schema.
- Implemented automated background jobs and **RESTful APIs** to update, retrieve, and validate user data.
- Wrote unit and integration tests with **Jest** covering API routes, scheduled jobs, and database interactions.

## SKILLS

**Languages:** Python, R, Bash, SQL, JavaScript, Java, C++, Assembly

**ML and Data Science:** PyTorch, Tensorflow, NumPy, Pandas, Scikit-Learn, NLTK, Seaborn, Tidyverse

**Bioinformatics:** Seurat, Scanpy, Bioconductor, Nextflow, SLURM, Samtools, BLAST, Docker, Conda, Git

**Web and Software Dev:** Flask, React.js, AWS (EC2, S3), MongoDB, RESTful APIs, HTML/CSS, GitHub Actions, Jira

**Awards:** USACO Gold Division, Dean's List (Multiple Semesters), Best Environmental Hack @ AngelHacks